

MySQL Shell 8.0.46

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Type 'help' or '?' for help; \quit to exit.

MySQL JS > \connect root@localhost

Creating a session to 'root@localhost'

Fetching schema names for auto-completion... Press ^C to stop.

Your MySQL connection id is 8 (X protocol)

Server version: 8.0.46 MySQL Community Server - GPL

No default schema selected; type \use <schema> to set one.

MySQL localhost:33060+ ssl JS > \sql

Switching to SQL mode... Commands end with ;

Fetching global names for auto-completion... Press ^C to stop.

MySQL localhost:33060+ ssl SQL > SHOW DATABASES;

```
+-----+
| Database          |
+-----+
| information_schema |
| my_first_db        |
| mysql              |
| performance_schema |
| sakila              |
| sys                 |
| unilabon            |
| world              |
| yousuf              |
+-----+
```

9 rows in set (0.0222 sec)

MySQL localhost:33060+ ssl SQL > create database lab_dbms;

Query OK, 1 row affected (0.0469 sec)

MySQL localhost:33060+ ssl SQL > use database lab_dbms;

Incorrect number of arguments for use command, usage:

\use <schema_name>

MySQL localhost:33060+ ssl SQL > use lab_dbms;

Default schema set to 'lab_dbms'.

Fetching global names, object names from 'lab_dbms' for auto-completion... Press ^C to stop.

MySQL localhost:33060+ ssl lab_dbms SQL > USE lab_dbms;

Default schema set to 'lab_dbms'.

Fetching global names, object names from 'lab_dbms' for auto-completion... Press ^C to stop.

MySQL localhost:33060+ ssl lab_dbms SQL > create table Student(SID INT PRIMARY KEY,Semester VARCHAR(50),CGPA float,Phone VARCHAR(15),Tution_Fee decimal);

Query OK, 0 rows affected (0.1085 sec)

MySQL localhost:33060+ ssl lab_dbms SQL > INSERT INTO Student (SID, Semester, CGPA, Phone, Tution_Fee)

-> VALUES

-> (101, '1st', 3.75, '01711111111', 25000.00),

-> (102, '2nd', 3.90, '01822222222', 27000.00),

-> (103, '3rd', 3.65, '01933333333', 26000.00);

Query OK, 3 rows affected (0.0287 sec)

Records: 3 Duplicates: 0 Warnings: 0

MySQL localhost:33060+ ssl lab_dbms SQL > decribe Student;

ERROR: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'decribe Student' at line 1

MySQL localhost:33060+ ssl lab_dbms SQL > SELECT * FROM Student;

```
+-----+-----+-----+-----+-----+
| SID | Semester | CGPA | Phone    | Tution_Fee |
+-----+-----+-----+-----+-----+
| 101 | 1st      | 3.75 | 01711111111 | 25000 |
| 102 | 2nd      | 3.9  | 01822222222 | 27000 |
| 103 | 3rd      | 3.65 | 01933333333 | 26000 |
+-----+-----+-----+-----+-----+
```

3 rows in set (0.0018 sec)

MySQL localhost:33060+ ssl lab_dbms SQL > DESCRIBE Student;

```
+-----+-----+-----+-----+-----+
| Field  | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
| SID    | int       | NO   | PRI | NULL    |      |
| Semester | varchar(50) | YES  |     | NULL    |      |
| CGPA    | float     | YES  |     | NULL    |      |
| Phone   | varchar(15) | YES  |     | NULL    |      |
| Tution_Fee | decimal(10,0) | YES  |     | NULL    |      |
+-----+-----+-----+-----+-----+
```

5 rows in set (0.0260 sec)

MySQL localhost:33060+ ssl lab_dbms SQL > DELETE From Student

-> ;

Query OK, 3 rows affected (0.0251 sec)

MySQL localhost:33060+ ssl lab_dbms SQL > INSERT INTO Student (SID, Semester, CGPA, Phone, Tution_Fee)

-> VALUES

-> (101, '1st', 3.75, '01711111111', 25000.00),

-> (102, '2nd', 3.90, '01822222222', 27000.00),

-> (103, '3rd', 3.65, '01933333333', 26000.00);

Query OK, 3 rows affected (0.0244 sec)

Records: 3 Duplicates: 0 Warnings: 0

MySQL localhost:33060+ ssl lab_dbms SQL > DESCRIBE Student;

```
+-----+-----+-----+-----+-----+
| Field  | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
```

SID	int	NO	PRI	NULL	
Semester	varchar(50)	YES		NULL	
CGPA	float	YES		NULL	
Phone	varchar(15)	YES		NULL	
Tution_Fee	decimal(10,0)	YES		NULL	

5 rows in set (0.0034 sec)

MySQL localhost:33060+ ssl lab_dbms SQL > ALTER TABLE Student
-> ADD Dept_id INT;

Query OK, 0 rows affected (0.0437 sec)

Records: 0 Duplicates: 0 Warnings: 0

MySQL localhost:33060+ ssl lab_dbms SQL > DESCRIBE Student;

Field	Type	Null	Key	Default	Extra
SID	int	NO	PRI	NULL	
Semester	varchar(50)	YES		NULL	
CGPA	float	YES		NULL	
Phone	varchar(15)	YES		NULL	
Tution_Fee	decimal(10,0)	YES		NULL	
Dept_id	int	YES		NULL	

6 rows in set (0.0040 sec)

MySQL localhost:33060+ ssl lab_dbms SQL > SELECT name AS na, CGPA AS GPA
-> FROM Student;

ERROR: 1054: Unknown column 'name' in 'field list'

MySQL localhost:33060+ ssl lab_dbms SQL >

MySQL localhost:33060+ ssl lab_dbms SQL > SELECT SID AS Student_ID,
-> CGPA AS GPA
-> FROM Student;

Student_ID	GPA
101	3.75
102	3.9
103	3.65

3 rows in set (0.0017 sec)

MySQL localhost:33060+ ssl lab_dbms SQL > SELECT Distinct Dept_id from Studnent;

ERROR: 1146: Table 'lab_dbms.studnent' doesn't exist

MySQL localhost:33060+ ssl lab_dbms SQL > SELECT DISTINCT Dept_id
-> FROM Student;

Dept_id
NULL

1 row in set (0.0018 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > UPDATE Student
      -> SET Dept_id = 1
      -> WHERE SID = 101;
```

Query OK, 1 row affected (0.0279 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
MySQL localhost:33060+ ssl lab_dbms SQL >
MySQL localhost:33060+ ssl lab_dbms SQL > UPDATE Student
      -> SET Dept_id = 2
      -> WHERE SID = 102;
```

Query OK, 1 row affected (0.0058 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
MySQL localhost:33060+ ssl lab_dbms SQL >
MySQL localhost:33060+ ssl lab_dbms SQL > UPDATE Student
      -> SET Dept_id = 3
      -> WHERE SID = 103;
```

Query OK, 1 row affected (0.0242 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
MySQL localhost:33060+ ssl lab_dbms SQL > decribe Student;
ERROR: 1064: You have an error in your SQL syntax; check the manual that corresponds to
your MySQL server version for the right syntax to use near 'decribe Student' at line 1
MySQL localhost:33060+ ssl lab_dbms SQL > DESCRIBE Student;
```

Field	Type	Null	Key	Default	Extra
SID	int	NO	PRI	NULL	
Semester	varchar(50)	YES		NULL	
CGPA	float	YES		NULL	
Phone	varchar(15)	YES		NULL	
Tution_Fee	decimal(10,0)	YES		NULL	
Dept_id	int	YES		NULL	

6 rows in set (0.0040 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT * FROM Student
      -> Where NOT dept_id = '1';
```

SID	Semester	CGPA	Phone	Tution_Fee	Dept_id
102	2nd	3.9	01822222222	27000	2
103	3rd	3.65	01933333333	26000	3

2 rows in set (0.0045 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT * FROM Student
      -> Where Tution_Fee BETWEEN 50000 AND 26000;
```

Empty set (0.0204 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT * FROM Student
->                                -> Where Tution_Fee BETWEEN
5000 AND 27000;
```

ERROR: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '-> Where Tution_Fee BETWEEN 5000 AND 27000' at line 2

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT * FROM Student
-> Where Tution_Fee BETWEEN 5000 AND 26000;
```

```
+----+-----+-----+-----+-----+-----+
| SID | Semester | CGPA | Phone    | Tution_Fee | Dept_id |
+----+-----+-----+-----+-----+-----+
| 101 | 1st      | 3.75 | 0171111111 | 25000 | 1 |
| 103 | 3rd      | 3.65 | 0193333333 | 26000 | 3 |
+----+-----+-----+-----+-----+-----+
```

2 rows in set (0.0016 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT * FROM Student
-> Where SID IN(101,102,103);
```

```
+----+-----+-----+-----+-----+-----+
| SID | Semester | CGPA | Phone    | Tution_Fee | Dept_id |
+----+-----+-----+-----+-----+-----+
| 101 | 1st      | 3.75 | 0171111111 | 25000 | 1 |
| 102 | 2nd      | 3.9  | 0182222222 | 27000 | 2 |
| 103 | 3rd      | 3.65 | 0193333333 | 26000 | 3 |
+----+-----+-----+-----+-----+-----+
```

3 rows in set (0.0013 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT * FROM Student
-> Where ;
```

ERROR: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '"' at line 2

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT * FROM Student
-> WHERE SID LIKE '1%';
```

```
+----+-----+-----+-----+-----+-----+
| SID | Semester | CGPA | Phone    | Tution_Fee | Dept_id |
+----+-----+-----+-----+-----+-----+
| 101 | 1st      | 3.75 | 0171111111 | 25000 | 1 |
| 102 | 2nd      | 3.9  | 0182222222 | 27000 | 2 |
| 103 | 3rd      | 3.65 | 0193333333 | 26000 | 3 |
+----+-----+-----+-----+-----+-----+
```

3 rows in set (0.0211 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT * FROM Student
-> WHERE SID LIKE '%1';
```

```
+----+-----+-----+-----+-----+-----+
| SID | Semester | CGPA | Phone    | Tution_Fee | Dept_id |
+----+-----+-----+-----+-----+-----+
| 101 | 1st      | 3.75 | 0171111111 | 25000 | 1 |
+----+-----+-----+-----+-----+-----+
```

1 row in set (0.0023 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT * FROM Student
-> WHERE SID NOT LIKE '2%';
```

```
+-----+-----+-----+-----+-----+
| SID | Semester | CGPA | Phone      | Tution_Fee | Dept_id |
+-----+-----+-----+-----+-----+
| 101 | 1st      | 3.75 | 0171111111 | 25000       | 1       |
| 102 | 2nd      | 3.9  | 0182222222 | 27000       | 2       |
| 103 | 3rd      | 3.65 | 0193333333 | 26000       | 3       |
+-----+-----+-----+-----+-----+
```

3 rows in set (0.0018 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT * FROM Student
-> WHERE SID NOT LIKE '%2';
```

```
+-----+-----+-----+-----+-----+
| SID | Semester | CGPA | Phone      | Tution_Fee | Dept_id |
+-----+-----+-----+-----+-----+
| 101 | 1st      | 3.75 | 0171111111 | 25000       | 1       |
| 103 | 3rd      | 3.65 | 0193333333 | 26000       | 3       |
+-----+-----+-----+-----+-----+
```

2 rows in set (0.0021 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT FROM Student
-> WHERE Phone IS NULL;
```

ERROR: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'FROM Student WHERE Phone IS NULL' at line 1

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT FROM Student
-> ;
```

ERROR: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'FROM Student' at line 1

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT *
-> FROM Student
-> WHERE Phone IS NULL;
```

Empty set (0.0017 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT FROM Student
-> LIMIT 1;
```

ERROR: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'FROM Student LIMIT 1' at line 1

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT *
-> FROM Student
-> LIMIT 1;
```

```
+-----+-----+-----+-----+-----+
| SID | Semester | CGPA | Phone      | Tution_Fee | Dept_id |
+-----+-----+-----+-----+-----+
| 101 | 1st      | 3.75 | 0171111111 | 25000       | 1       |
+-----+-----+-----+-----+-----+
```

1 row in set (0.0218 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT COUNT(*)
-> FROM Student;
```

```
+-----+
| COUNT(*) |
+-----+
|      3 |
+-----+
```

1 row in set (0.0314 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT COUNT(Phone)
                                -> FROM Student;
```

ERROR: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'Student' at line 2

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT COUNT(Phone)
                                -> FROM Student;
```

```
+-----+
| COUNT(Phone) |
+-----+
|          3 |
+-----+
```

1 row in set (0.0014 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT SUM(Tuion_Fee)
                                -> FROM Student;
```

ERROR: 1054: Unknown column 'Tuion_Fee' in 'field list'

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT SUM(Tuition_Fee)
                                -> FROM Student;
```

ERROR: 1054: Unknown column 'Tuition_Fee' in 'field list'

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT SUM(Tuition_Fee)
                                -> SELECT SUM(Tution_Fee) FROM Student;
```

ERROR: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'SELECT SUM(Tution_Fee) FROM Student' at line 2

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT SUM(Tution_Fee)
                                -> FROM Student;
```

```
+-----+
| SUM(Tution_Fee) |
+-----+
|          78000 |
+-----+
```

1 row in set (0.0016 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT SUM(Tution_Fee),
                                -> Avg(Tution_Fee),
                                -> MINI (CGPA)
                                -> ,
                                -> MAX(CGPA)
                                -> COUNT(SID)
                                -> FROM Student;
```

ERROR: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'COUNT(SID) FROM Student' at line 6

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT
```

```

-> SUM(Tution_Fee),
-> AVG(Tution_Fee),
-> MIN(CGPA),
-> MAX(CGPA),
-> COUNT(SID)
-> FROM Student;

```

```

+-----+-----+-----+-----+-----+
| SUM(Tution_Fee) | AVG(Tution_Fee) | MIN(CGPA) | MAX(CGPA) | COUNT(SID) |
+-----+-----+-----+-----+-----+
|      78000 | 26000.0000 | 3.65 | 3.9 | 3 |
+-----+-----+-----+-----+-----+

```

1 row in set (0.0026 sec)

MySQL localhost:33060+ ssl lab_dbms SQL > SELECT

```

-> COUNT(*),
-> ROUND(AVG(CGPA), 2),
-> MAX(CGPA),
-> MIN(CGPA)
-> FROM Student;

```

```

+-----+-----+-----+-----+
| COUNT(*) | ROUND(AVG(CGPA), 2) | MAX(CGPA) | MIN(CGPA) |
+-----+-----+-----+-----+
|      3 | 3.77 | 3.9 | 3.65 |
+-----+-----+-----+-----+

```

1 row in set (0.0205 sec)

MySQL localhost:33060+ ssl lab_dbms SQL > SELECT

```

-> COUNT(*) AS total_students,
-> ROUND(AVG(CGPA), 2) AS average_cgpa,
-> MAX(CGPA) AS highest_cgpa,
-> MIN(CGPA) AS lowest_cgpa
-> FROM Student;

```

```

+-----+-----+-----+-----+
| total_students | average_cgpa | highest_cgpa | lowest_cgpa |
+-----+-----+-----+-----+
|      3 | 3.77 | 3.9 | 3.65 |
+-----+-----+-----+-----+

```

1 row in set (0.0021 sec)

MySQL localhost:33060+ ssl lab_dbms SQL > SELECT SID,CGPA

```

-> FROM Student
-> ORDER BY CGPA,DESC;

```

ERROR: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'DESC' at line 3

MySQL localhost:33060+ ssl lab_dbms SQL > SELECT SID, CGPA

```

-> FROM Student
-> ORDER BY CGPA DESC;

```

```

+----+----+
| SID | CGPA |
+----+----+
| 102 | 3.9 |

```

```
| 101 | 3.75 |
| 103 | 3.65 |
```

```
+-----+-----+
```

3 rows in set (0.0013 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > SELECT SID, CGPA
      -> FROM Student
      -> ORDER BY CGPA ASC;
```

```
+-----+-----+
```

```
| SID | CGPA |
```

```
+-----+-----+
```

```
| 103 | 3.65 |
```

```
| 101 | 3.75 |
```

```
| 102 | 3.9 |
```

```
+-----+-----+
```

3 rows in set (0.0010 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > DESCRIBE Student;
```

```
+-----+-----+-----+-----+-----+
```

```
| Field      | Type      | Null | Key | Default | Extra |
```

```
+-----+-----+-----+-----+-----+
```

```
| SID        | int       | NO   | PRI | NULL    |      |
```

```
| Semester   | varchar(50) | YES  |     | NULL    |      |
```

```
| CGPA       | float     | YES  |     | NULL    |      |
```

```
| Phone      | varchar(15) | YES  |     | NULL    |      |
```

```
| Tution_Fee | decimal(10,0) | YES  |     | NULL    |      |
```

```
| Dept_id    | int       | YES  |     | NULL    |      |
```

```
+-----+-----+-----+-----+-----+
```

6 rows in set (0.0114 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > ALTER TABLE Student
      -> ADD Department VARCHAR(50);
```

Query OK, 0 rows affected (0.0597 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
MySQL localhost:33060+ ssl lab_dbms SQL > DESCRIBE Student;
```

```
+-----+-----+-----+-----+-----+
```

```
| Field      | Type      | Null | Key | Default | Extra |
```

```
+-----+-----+-----+-----+-----+
```

```
| SID        | int       | NO   | PRI | NULL    |      |
```

```
| Semester   | varchar(50) | YES  |     | NULL    |      |
```

```
| CGPA       | float     | YES  |     | NULL    |      |
```

```
| Phone      | varchar(15) | YES  |     | NULL    |      |
```

```
| Tution_Fee | decimal(10,0) | YES  |     | NULL    |      |
```

```
| Dept_id    | int       | YES  |     | NULL    |      |
```

```
| Department | varchar(50) | YES  |     | NULL    |      |
```

```
+-----+-----+-----+-----+-----+
```

7 rows in set (0.0043 sec)

```
MySQL localhost:33060+ ssl lab_dbms SQL > UPDATE Student
      -> SET Department = 'CSE'
      -> WHERE SID = 101;
```

Query OK, 1 row affected (0.0052 sec)

Rows matched: 1 Changed: 1 Warnings: 0

MySQL localhost:33060+ ssl lab_dbms SQL >

MySQL localhost:33060+ ssl lab_dbms SQL > UPDATE Student

-> SET Department = 'EEE'

-> WHERE SID = 102;

Query OK, 1 row affected (0.0068 sec)

Rows matched: 1 Changed: 1 Warnings: 0

MySQL localhost:33060+ ssl lab_dbms SQL >

MySQL localhost:33060+ ssl lab_dbms SQL > UPDATE Student

-> SET Department = 'BBA'

-> WHERE SID = 103;

Query OK, 1 row affected (0.0236 sec)

Rows matched: 1 Changed: 1 Warnings: 0

MySQL localhost:33060+ ssl lab_dbms SQL > DESCRIBE Student;

Field	Type	Null	Key	Default	Extra
SID	int	NO	PRI	NULL	
Semester	varchar(50)	YES		NULL	
CGPA	float	YES		NULL	
Phone	varchar(15)	YES		NULL	
Tution_Fee	decimal(10,0)	YES		NULL	
Dept_id	int	YES		NULL	
Department	varchar(50)	YES		NULL	

7 rows in set (0.0038 sec)

MySQL localhost:33060+ ssl lab_dbms SQL > SELECT * FROM Student;

SID	Semester	CGPA	Phone	Tution_Fee	Dept_id	Department
101	1st	3.75	01711111111	25000	1	CSE
102	2nd	3.9	01822222222	27000	2	EEE
103	3rd	3.65	01933333333	26000	3	BBA

3 rows in set (0.0013 sec)

MySQL localhost:33060+ ssl lab_dbms SQL > SELECT Dept_id,COUNT(Dept_id)

-> FROM Student

-> ;

ERROR: 1140: In aggregated query without GROUP BY, expression #1 of SELECT list contains nonaggregated column 'lab_dbms.Student.Dept_id'; this is incompatible with sql_mode=only_full_group_by

MySQL localhost:33060+ ssl lab_dbms SQL > SELECT Department, COUNT(*)

-> FROM Student

-> GROUP BY Department;

```
+-----+-----+
| Department | COUNT(*) |
+-----+-----+
| CSE      |      1 |
| EEE      |      1 |
| BBA      |      1 |
+-----+-----+
```

3 rows in set (0.0237 sec)

MySQL localhost:33060+ ssl lab_dbms SQL >